

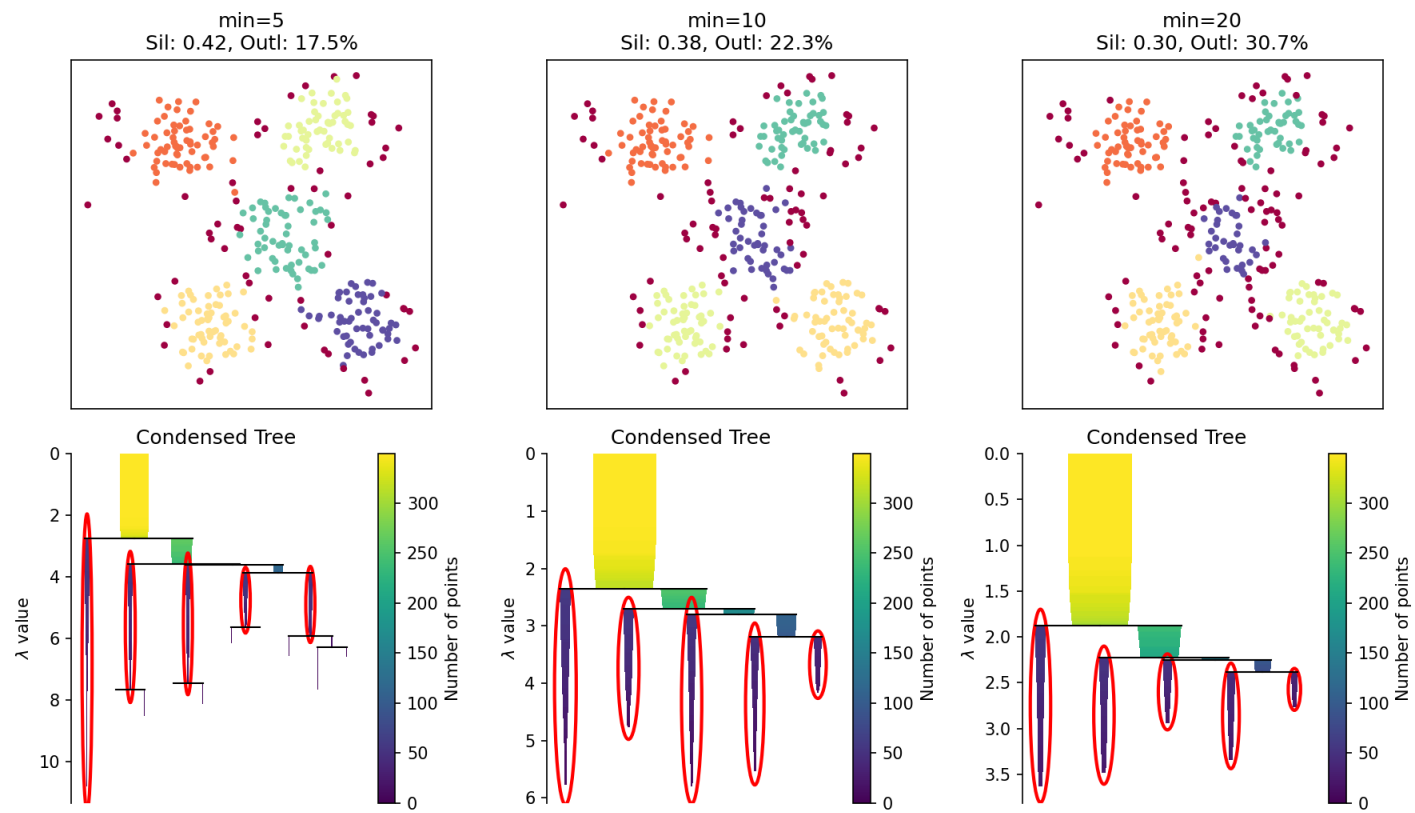


HDBSCAN

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Curso	Machine Learning 2
Unidade curricular	Computer Science
Ano letivo	Clique para introduzir o ano letivo
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Data	12/05/2025

Dataset 1

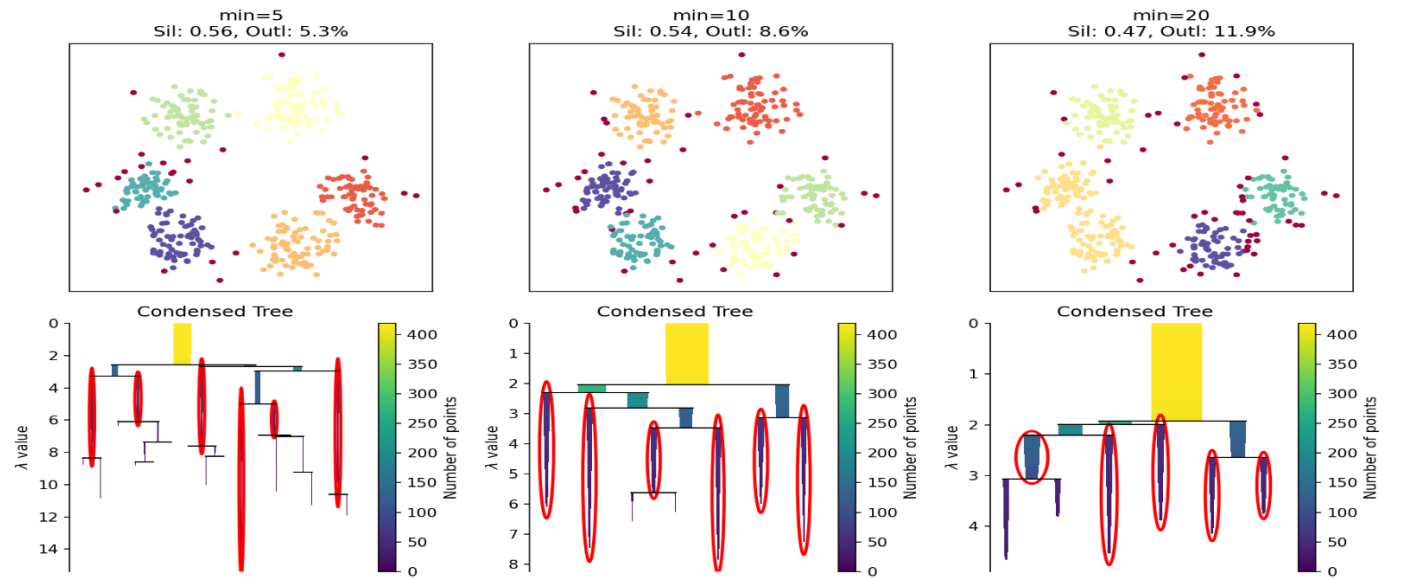
dataset1 - HDBSCAN Results



To determine the optimal minimum cluster size for HDBSCAN on dataset1, we tested values of 5, 10, and 20. Setting minimum cluster size = 5 resulted in the best performance, achieving the highest silhouette score (0.42) and the lowest outlier percentage (17.5%). This indicates that the clusters were compact and well separated. As minimum cluster size increased, the algorithm became more conservative, rejecting more points as noise (over 30% at min=20) and reducing the clarity of the cluster structure. Therefore, we chose minimum cluster size = 5 as the optimal value, as it offered the best compromise between cluster quality, completeness, and visual separation.

Dataset 2

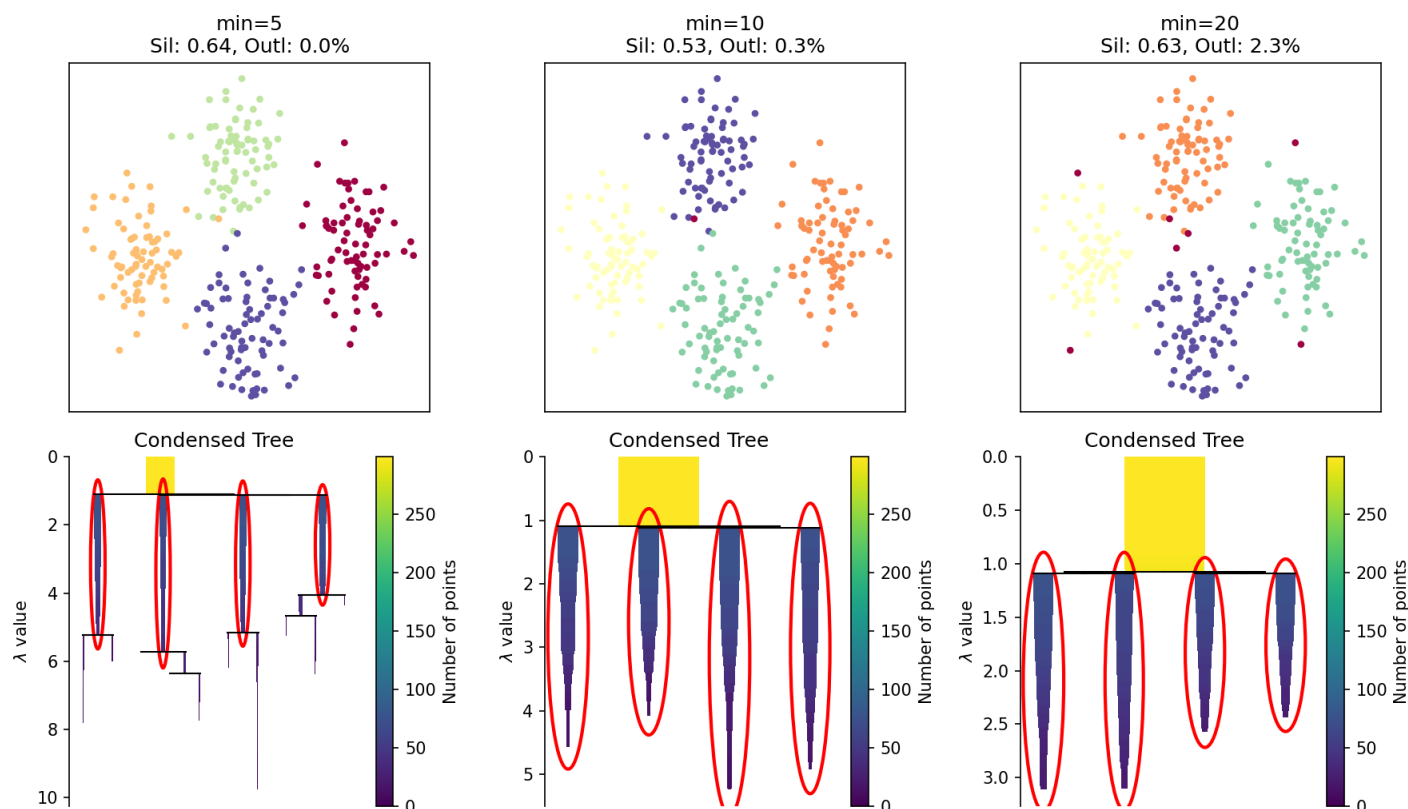
dataset2 - HDBSCAN Results



For dataset2, we tested three values for the minimum cluster size parameter in the HDBSCAN algorithm: 5, 10, and 20. The best results were achieved with a minimum cluster size of 5, which produced the highest silhouette score (0.56) and the lowest percentage of outliers (5.3%). Although a value of 10 also performed well (silhouette: 0.54), it resulted in slightly more outliers and less compact clusters. Increasing the minimum cluster size to 20 further reduced the cluster quality, with a lower silhouette score (0.47) and more noise (11.9% outliers). Therefore, a minimum cluster size of 5 is the optimal choice, as it maintains clean and well-separated clusters while minimizing data loss.

Dataset 3

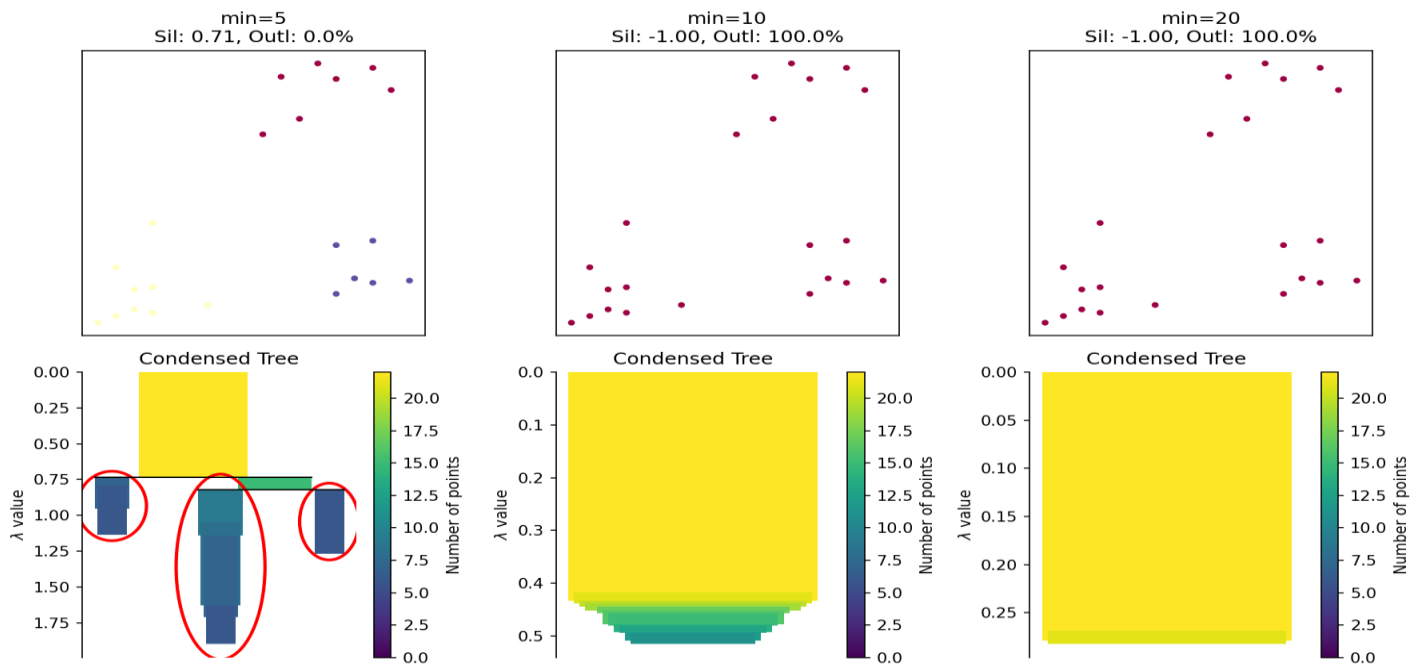
dataset3 - HDBSCAN Results



For dataset 3, we evaluated the HDBSCAN algorithm using minimum cluster size values of 5, 10, and 20. The configuration with a minimum cluster size of 5 achieved the highest silhouette score of 0.64 and no outliers (0.0%), indicating excellent separation and compact clustering. A minimum cluster size of 20 produced a very similar silhouette score (0.63) but introduced a small number of outliers (2.3%), while setting 10 had a worse overall performance, with a lower silhouette score of 0.53. Although all configurations visually detect the same underlying structure, the configuration with a minimum cluster size of 5 stands out as the optimal choice due to its perfect preservation of all data and the highest clustering quality.

Dataset 4

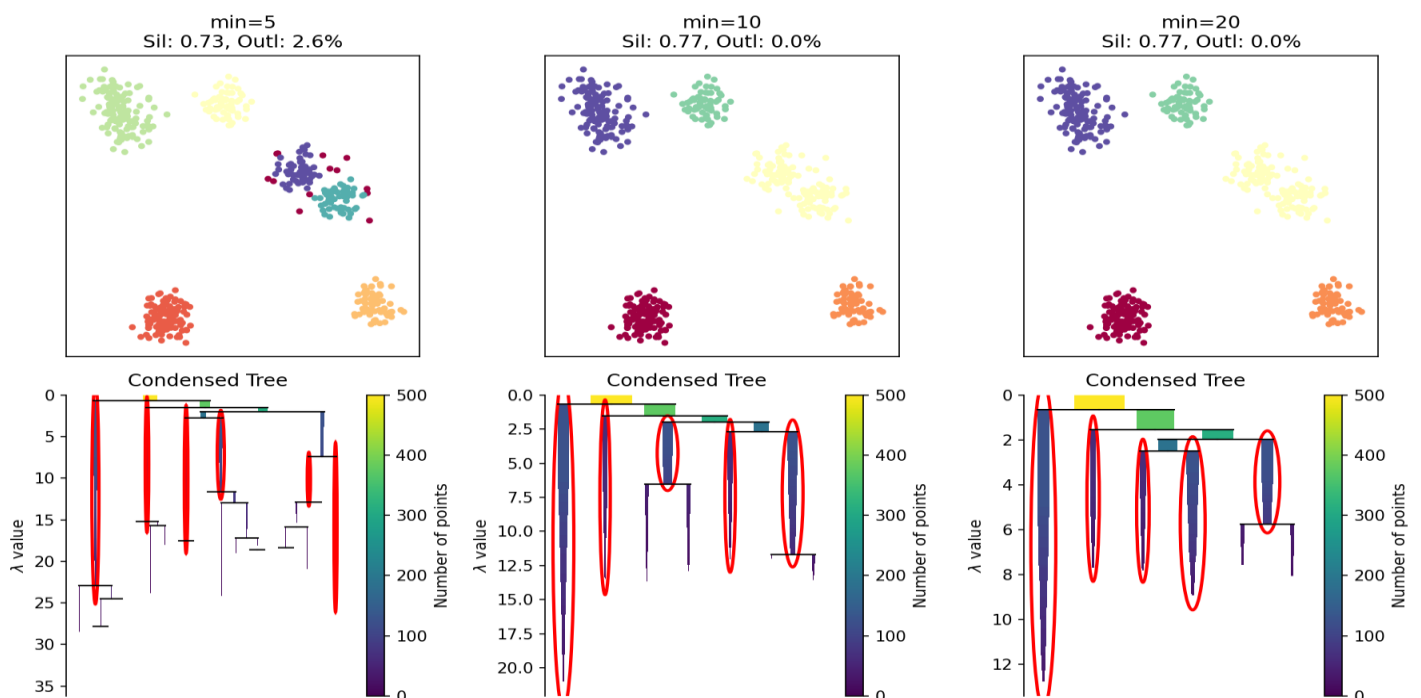
dataset4 - HDBSCAN Results



On dataset4, we tested HDBSCAN using minimum cluster size values of 5, 10, and 20. The best result was achieved with a minimum cluster size of 5, which yielded a high silhouette score of 0.71 and 0% outliers, indicating a clean and meaningful clustering structure. When the minimum cluster size was increased to 10 or 20, the algorithm failed to detect any clusters: all data points were labeled as noise (100% outliers) and the silhouette score decreased to -1.00. This suggests that the dataset contains a very small number of dense clusters, and larger minimum cluster sizes are too stringent to capture them. Therefore, a minimum cluster size of 5 is clearly the optimal choice, as it successfully identifies all clusters without discarding any data.

Dataset 5 Blobs_8

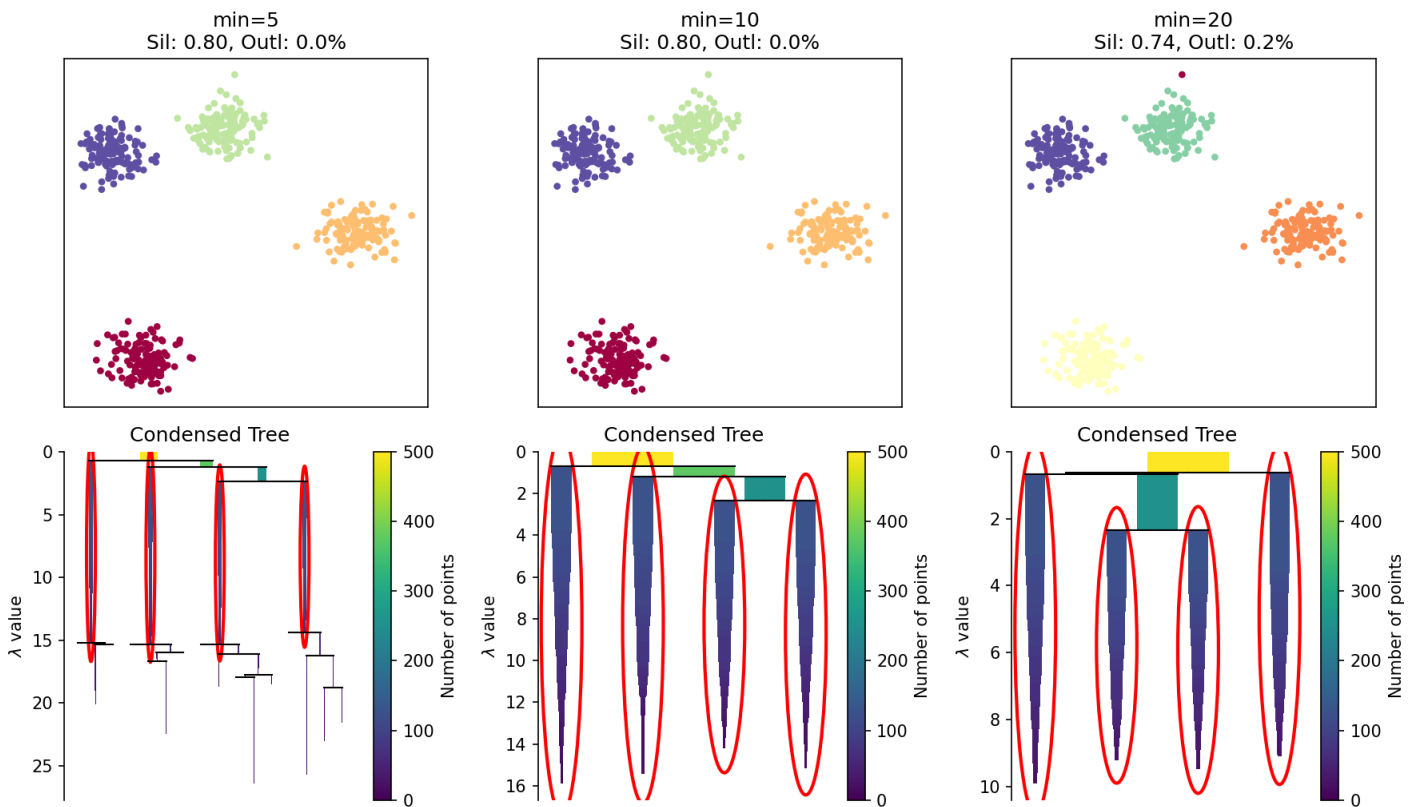
Blobs_8_centers - HDBSCAN Results



For the synthetic Blobs_8_centers dataset, HDBSCAN was tested with minimum cluster size values of 5, 10, and 20. The best performance was observed at both 10 and 20, with a silhouette score of 0.77 and 0% outliers, indicating clean, well-separated clusters with perfect data preservation. The result for minimum cluster size 5 was slightly less optimal, with a silhouette score of 0.73 and a small number of outliers (2.6%). This suggests that although HDBSCAN was effective in all settings, larger minimum cluster sizes (10 and 20) helped to enhance the integrity of the clusters, while eliminating noise. Therefore, the optimal configuration is either 10 or 20, with a slight preference for minimum cluster size = 10 due to the balance between granularity and stability.

Dataset 6 Blobs_4

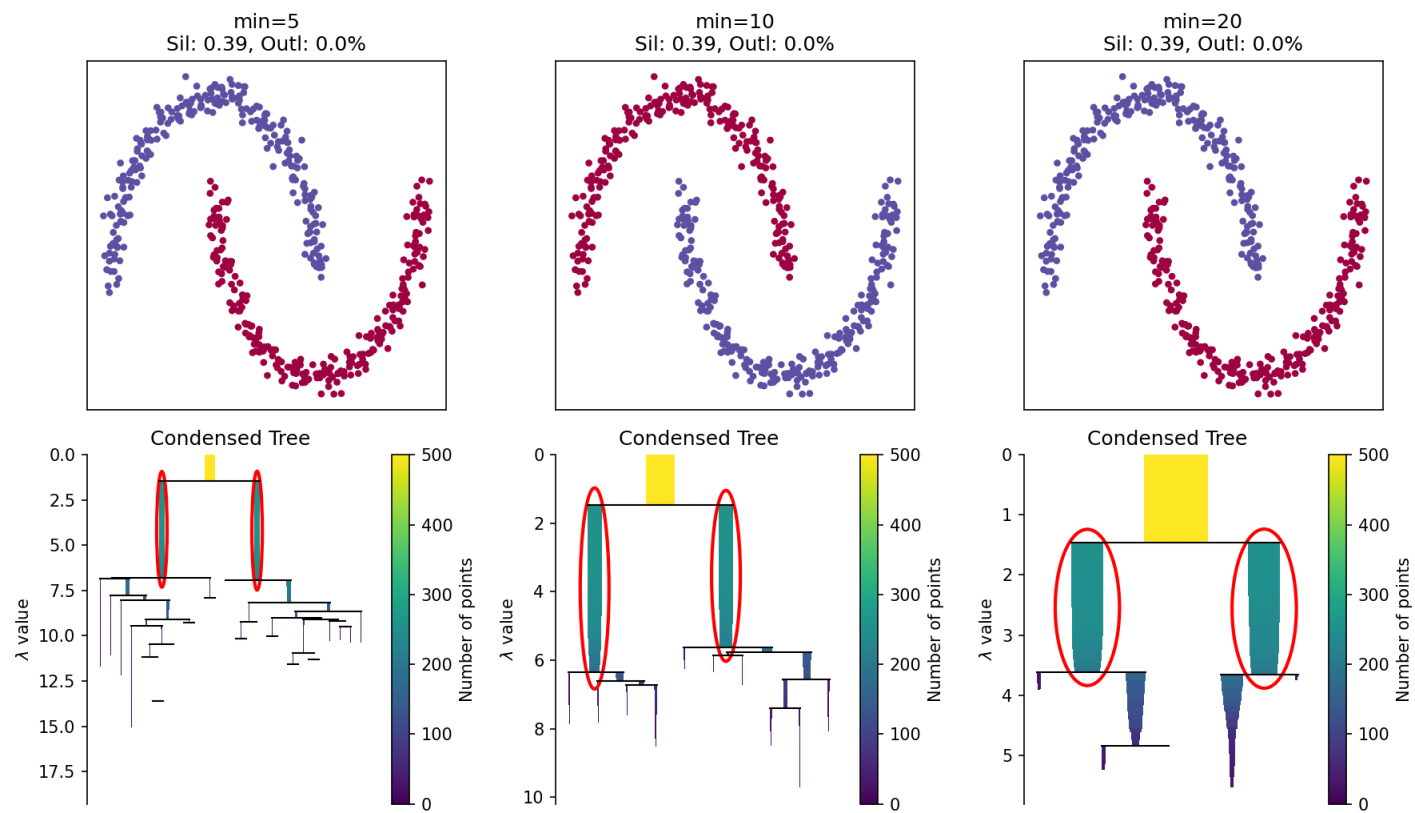
Blobs_4_centers - HDBSCAN Results



On the Blobs_4_centers dataset, HDBSCAN was applied with minimum cluster size values of 5, 10, and 20. Both 5 and 10 produced identical silhouette scores of 0.80 and 0% outliers, indicating a perfectly clean and well-separated clustering result. When the minimum cluster size was increased to 20, the silhouette score decreased slightly to 0.74 and a few outliers (0.2%) appeared. Visually, all configurations identified the same four dense clusters. Since both 5 and 10 perform equally well in terms of clustering quality and completeness, the optimal choice may be either — but minimum cluster size = 10 is preferable for additional robustness, especially on larger or noisier datasets.

Dataset 7 Moons

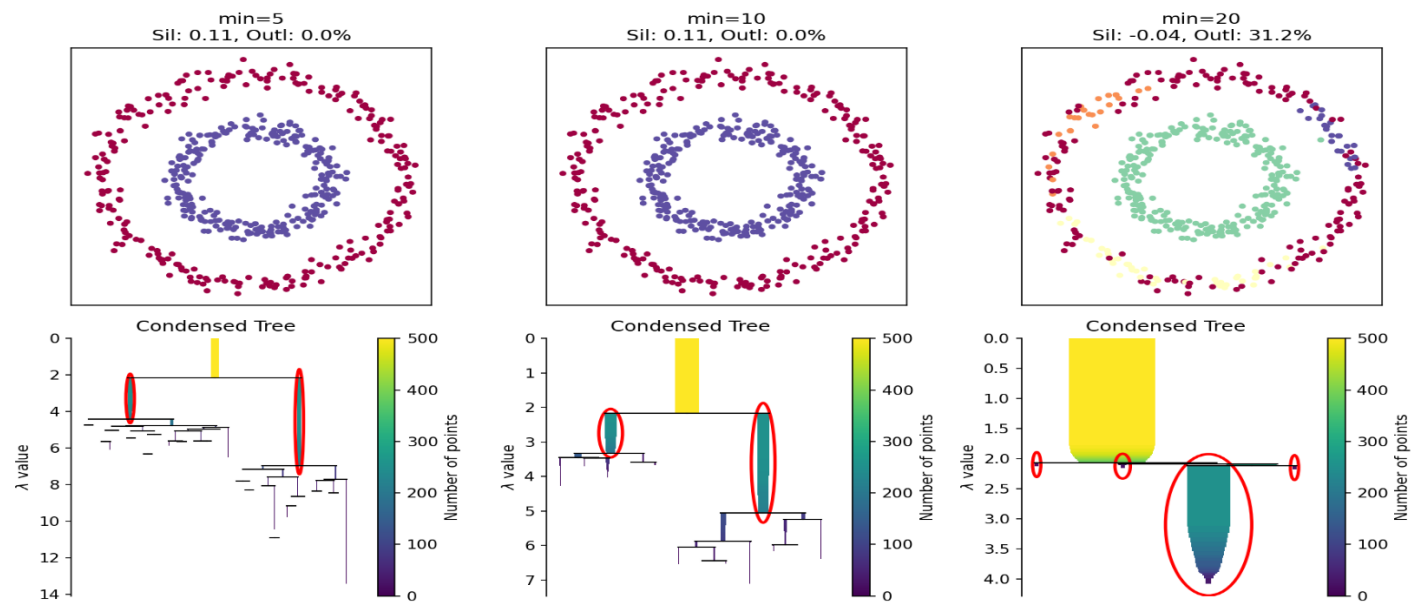
Moons - HDBSCAN Results



On the Moons dataset, HDBSCAN was evaluated using minimum cluster size values of 5, 10, and 20. Interestingly, all three settings produced identical silhouette scores of 0.39 and 0% outliers, indicating stable and consistent clustering performance. Despite the nonlinear structure of the moons, HDBSCAN successfully captured the two curved clusters without misclassifying points as noise. Since the results are almost identical across configurations, the choice of minimum cluster size does not significantly affect performance in this case. Therefore, any value between 5, 10, or 20 is appropriate, although a moderate setting such as 10 may provide slightly better robustness to more complex or noisy variations of the dataset.

Dataset 8 Circles

Circles - HDBSCAN Results



For the Circles dataset, HDBSCAN was evaluated using minimum cluster size values of 5, 10, and 20. Both 5 and 10 yielded the same silhouette score of 0.11 with 0% outliers, successfully capturing the two concentric circular clusters despite the low density between them. However, when the minimum cluster size was increased to 20, the performance decreased significantly: the silhouette score decreased to -0.04 and 31.2% of the data were flagged as outliers. This suggests that a larger minimum cluster size is too strict for this type of geometric structure, resulting in the model not correctly identifying the rings. Therefore, the best choice is minimum cluster size = 10, offering a clean, stable result without outliers, even if the overall separation of the clusters is inherently difficult due to the shape.

Dataset	Optimal K	Silhouette	Outliers
Dataset 1	5	0,42	17.5%
Dataset 2	5	0,56	5.3%
Dataset 3	5	0,64	0.0%
Dataset 4	5	0,71	0.0%
Blobs 4	10	0,80	0.0%
Blobs 8	10	0,77	0.0%
Circles	10	0,11	0.0%
Moons	10	0,39	0.0%

